Part 1 · Mathematics

How Grade 8 Math Is Built

The Grade 8 Mathematics test has **40 questions worth 48 points.** Every question is drawn from one of **four strands** (the state calls them reporting categories). Knowing the strands tells you exactly what to study — and this book teaches **every skill in every strand.**

Strand	What It Covers	On the Test
1 · Numerical Representations & Relationships	Real numbers, irrational numbers, scientific notation, ordering numbers	~4 questions
2 · Computations & Algebraic Relationships	Slope, linear equations, functions, one-variable equations	~15 questions (largest)
3 · Geometry & Measurement	Dilations, transformations, volume, surface area, Pythagorean theorem, angles	~13 questions
4 · Data Analysis & Personal Financial Literacy	Scatterplots, trend lines, mean absolute deviation, interest, saving	~8 questions

Readiness vs. Supporting Standards

- ★ **Readiness standard** — tested heavily every year; carries most of the points. Master these first.
- ○ **Supporting standard** — tested less often, but still fair game.

The Process Standards (8.1 A–G)

STAAR also weaves in **mathematical process standards**: applying math to real situations, choosing a strategy, using tools, and explaining your reasoning. You practice these inside every lesson and test — they are *how* you do the math, not a separate topic.

> **Key Idea:** Read each lesson, do the warm-up drills, then prove it on the practice tests. Use the **reference sheet** on the next page the whole way through — it is yours on the real test, too.

Grade 8 Mathematics Reference Sheet

On the real STAAR test, these formulas are provided for you. Learn where each one is so you can find it fast. (This replicates the key formulas from the STAAR Grade 8 Mathematics Reference Materials.)

Perimeter & Circumference	**Area**
Circle: $\mathbf{C = 2\pi r}$ or $\mathbf{C = \pi d}$	Triangle: $\mathbf{A = \frac{1}{2}bh}$ Rectangle / Parallelogram: $\mathbf{A = bh}$ Trapezoid: $\mathbf{A = \frac{1}{2}(b_1 + b_2)h}$ Circle: $\mathbf{A = \pi r^2}$
Surface Area (B = base area, P = base perimeter)	**Volume** (B = base area)
Prism — Lateral: $\mathbf{S = Ph}$ · Total: $\mathbf{S = Ph + 2B}$ Cylinder — Lateral: $\mathbf{S = 2\pi rh}$ · Total: $\mathbf{S = 2\pi rh + 2\pi r^2}$	Prism / Cylinder: $\mathbf{V = Bh}$ Cone: $\mathbf{V = \frac{1}{3}Bh}$ Sphere: $\mathbf{V = \frac{4}{3}\pi r^3}$
Algebra & Graphing	**Personal Financial Literacy**
Slope: $\mathbf{m = (y_2 - y_1) / (x_2 - x_1)}$ Slope-intercept form: $\mathbf{y = mx + b}$ Direct variation: $\mathbf{y = kx}$ Pythagorean theorem: $\mathbf{a^2 + b^2 = c^2}$	Simple interest: $\mathbf{I = Prt}$ Compound interest: $\mathbf{A = P(1 + r/n)^{nt}}$
Useful Constants & Conversions	
$\mathbf{\pi \approx 3.14}$ (or $22/7$) · Metric: 1 m = 100 cm = 1000 mm; 1 km = 1000 m · Customary: 1 ft = 12 in; 1 yd = 3 ft; 1 mi = 5,280 ft	

Strategy: You do **not** need to memorize these — but you must know they exist and how to use them. Every worked example that uses a formula shows you which one and why.

Strand 1 · Numerical Representations and Relationships

What this strand tests: how to represent, compare, and order real numbers — including irrational numbers and scientific notation.

Standards covered:

- ○ **8.2A** — Describe relationships between sets of real numbers using a visual representation.
- ○ **8.2B** — Approximate an irrational number (including π and square roots of numbers less than 225) and locate it on a number line.
- ○ **8.2C** — Convert between standard decimal notation and scientific notation.
- ★ **8.2D** — Order a set of real numbers from mathematical and real-world contexts. *(Readiness)*

8.2A · The Real Number System ○

Every number you use at Grade 8 is a **real number.** Real numbers divide into groups nested inside one another:

- **Natural:** 1, 2, 3, … **Whole:** 0, 1, 2, 3, … **Integers:** … −2, −1, 0, 1, 2, …
- **Rational:** any number written as a fraction a/b — integers, terminating decimals (0.75), repeating decimals (0.333…).
- **Irrational:** cannot be written as a simple fraction; decimals never end or repeat — such as **π** and **√2**.

> **Key Idea:** The groups are *nested* — every natural number is also whole, an integer, and rational. But rational and irrational never overlap.

Worked example. Classify √16 and √20. √16 = 4 → natural, whole, integer, **rational.** √20 ≈ 4.472… → **irrational.**

Watch Out: A square root is irrational only when the number is **not** a perfect square. √25 = 5 is rational; √26 is irrational.

Try It. Every integer is a rational number — true or false? *(True: any integer n = n/1.)*

8.2B · Approximating Irrational Numbers ○

To estimate a square root, find the two **perfect squares** it falls between.

Worked example. Approximate √50. $49 = 7^2$ and $64 = 8^2$, so √50 is between **7 and 8**; just above 49, so √50 ≈ **7.1** — mark it just right of 7 on a number line. **π ≈ 3.14**, between 3 and 4.

Watch Out: "Less than 225" means the number under the root is below 225 — roots up to about √224 ≈ 14.97.

Try It. Between which two whole numbers is √30? *(5 and 6, since 25 < 30 < 36.)*

8.2C · Scientific Notation ○

Scientific notation writes a number as $\mathbf{a \times 10^n}$, where $\mathbf{1 \le |a| < 10}$ and n is an integer. Large numbers → positive exponent; small numbers (less than 1) → negative exponent.

- 4,500,000 = $\mathbf{4.5 \times 10^6}$ · 0.00032 = $\mathbf{3.2 \times 10^{-4}}$ · 7.1×10^5 = **710,000**

Watch Out: The factor *a* must be at least 1 and less than 10. Writing 45×10^5 is wrong — fix to 4.5×10^6.

Try It. Write 0.0067 in scientific notation. *(6.7×10^{-3}.)*

8.2D · Ordering Real Numbers ★ (Readiness)

Numbers often appear in different forms at once — fractions, decimals, percents, roots, scientific notation. **Convert them all to decimals, then compare.**

Worked example. Order from least to greatest: ⅗, 0.58, √0.36, 62%.
⅗ = 0.60 · 0.58 = 0.58 · √0.36 = 0.60 · 62% = 0.62 → **0.58, then ⅗ = √0.36, then 62%.**

> **Key Idea:** One common form — usually the decimal — lets you compare very different-looking numbers fairly. A number line helps you *see* the order.

Watch Out: On a negative number line, the number farther **left** is **smaller**. −0.8 < −0.3.

Try It. Order greatest to least: 0.7, ⅔, 71%. *(71% = 0.71, 0.7, ⅔ ≈ 0.667.)*

Strand 1 Checklist

- ☐ 8.2A Classify real numbers and describe how the sets relate
- ☐ 8.2B Approximate irrational numbers and place them on a number line
- ☐ 8.2C Convert between standard and scientific notation
- ☐ 8.2D Order a mixed set of real numbers

Ready to practice? Warm-up drills for Strand 1 come after all four strand lessons.